## Design Project: Stepped Transmission Shaft Ø60×200mm

ACTIVE ESTIMATION

Part Type: SHAFT

Generated: 10/14/2026, 6:22:26 AM

Engineering Author: Viraj Bansod

Currency: INR

Machining / Waste Factor: 1.00x

## 1. Primary Part Material & Mass Metrics

| CALCULATED VOLUME | ESTIMATED MASS | RAW MATERIAL COST | TOTAL WITH OVERHEAD |
|---|---|---|---|
| 520.25 cm³ | 4.162 kg | INR 1,186.20 | INR 1,186.20 |

### Assigned Material Specifications

| | | | |
|---|---|---|---|
| **Material Name** | Stainless Steel 304 (AISI 304 / 1.4301) | **Young's Modulus (E)** | 193 GPa |
| **Material Category** | Ferrous | **Yield Strength (Ãy)** | 215 MPa |
| **Material Density** | 8 g/cm³ | **Base Cost per kg** | INR 285 / kg |
| **Machinability Rating** | 45 / 100 | **Thermal Conductivity** | 16.2 W/m·K |

## 2. Full Assembly Bill of Materials (BOM) Cost Rollup

| # | Component Name | Part Type | Assigned Material | Mass (kg) | Cost (INR) |
|---|---|---|---|---|---|
| 1 | Stepped Transmission Shaft | shaft | Stainless Steel 304 (A | 4.162 | **INR 1,186.20** |
| 2 | High-Torque Pinion Gear | gear | EN19 / 40Cr4Mo2 Chrome | 0.946 | **INR 156.10** |
| 3 | Deep Groove Radial Bearing | bearing | Structural Steel (ASTM | 0.453 | **INR 34.00** |
| | **TOTAL ASSEMBLY COST ROLLUP** | | | **5.561 kg** | **INR 1,376.30** |

### Batch Production Scale Projections

| | | | |
|---|---|---|---|
| Prototype (1 unit) | 0% (Standard) | Unit: INR 1,186.20 | **Total: INR 1,186.20** |
| Pilot Batch (10 units) | 8% Batch Discount | Unit: INR 1,091.30 | **Total: INR 10,913.04** |
| Production Run (100 units) | 18% Tooling Amortized | Unit: INR 972.68 | **Total: INR 97,268.40** |
| Mass Production (1000 units) | 28% Bulk Raw Material | Unit: INR 854.06 | **Total: INR 8,54,064.00** |

**ENGINEERING COST CERTIFICATION**

Calculations derived using analytical volumetric CAD primitives, standard alloy density indices, and current raw material pricing.

Certified by: CADVANTA Cost Estimation Engine | Viraj Bansod Mechanical Systems

`[VERIFIED: DOC-EST-MV7Q38IT]`